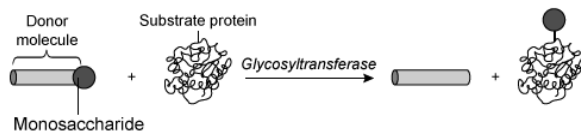


Based on the predicted structure of a certain gene product, the corresponding gene is believed to encode a glycosyltransferase that post-translationally modifies a 100 kDa substrate protein as shown in the diagram below.



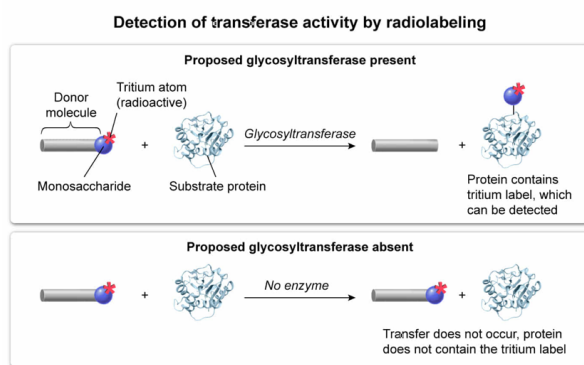
Which of the following experiments would best test this prediction?

- ☐ A. Fluorescently label the proposed glycosyltransferase, mix it with the substrate protein and donor molecule, then assess whether the proposed glycosyltransferase is retained on a column that binds the monosaccharide. [15%]
- ☐ B. Mix the substrate protein and donor molecule, with or without the proposed glycosyltransferase, then perform a Western blot using an antibody specific to the substrate protein. [26%]
- ☐ C. Incubate the donor molecule and substrate protein, with or without the proposed glycosyltransferase, then use mass spectrometry to measure the mass of the donor molecule. [17%]
- ☒ D. Add a tritium label to the monosaccharide, incubate the donor molecule and substrate protein with or without the proposed glycosyltransferase, then measure tritium levels in the purified substrate protein. [40%]

Correct

40% answered correctly

Explanation:



Transferases are a class of enzymes that **move a chemical group** from one molecule to another. For example, glycosyltransferases move carbohydrate (glycosyl) groups. To confirm that a given enzyme is a glycosyltransferase, the **movement of the carbohydrate** group (a monosaccharide in this case) from one molecule to another **must be observed** to a much greater degree **in the presence of the enzyme** than in its absence.

Chemical groups are commonly tracked by replacing one type of atom in the group with a **radioactive isotope** of that atom. Tritium is a radioactive isotope of hydrogen that is often used to track organic chemical groups. If the **monosaccharide portion** of the donor molecule is **labeled with tritium**, and the monosaccharide is subsequently transferred to a protein, then the **radioactivity of the tritium will be detected** in the purified protein. If this is observed when the proposed glycosyltransferase is present in the incubation but not when it is absent, this would provide strong evidence that the enzyme really is a glycosyltransferase.

(Choice A) **Fluorescently labeling** the *substrate protein* and then using the label to determine whether the substrate protein is retained on a column that binds the monosaccharide would show whether the monosaccharide has been transferred to the protein. However, labeling the proposed *glycosyltransferase* and measuring its binding to the column would *not* show whether monosaccharide was transferred.

(Choice B) A **Western blot** using an antibody against the substrate protein would show only that the protein is present, not whether it received the monosaccharide. Although post-translational modifications can cause gel mobility shifts, the mass of the substrate protein (100 kDa) is much greater than the mass of a monosaccharide, so the resulting mobility shift would likely be undetectable. For a Western blot to work, the antibody would have to specifically bind the *glycosylated form* of the protein.

(Choice C) Measuring the mass of the *substrate protein* could show that it has gained the mass of one monosaccharide, which would indicate that glycosylation has occurred. However, measuring the mass of the *donor molecule* would show only that the carbohydrate was removed from the donor molecule; it would not necessarily mean the carbohydrate was subsequently linked to the substrate protein.

Educational objective:

To conclude that a given enzyme is a transferase, the chemical group of interest must be observed moving from one molecule to another in the presence, but not the absence, of the enzyme.

Time spent:0 sec

Subject:
Biochemistry

QID:403384

System:
Enzymes